Single-cell eQTL-guided translational prioritization in PBC

A signal-aware evidence funnel from immune-cell regulation to a bounded therapeutic hypothesis

GENETICS → CONTEXT → TRANSLATION

1 Discovery

PBC GWAS × OneK1K sc-eQTL

56 loci • 17 immune-cell populations

16 overlap loci | 152 records

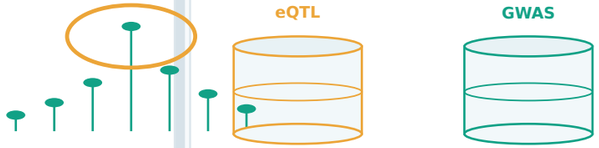
2 Signal resolution

Signed LD • SuSiE fine-mapping

Signal-level coloc.susie

10 loci | 23 formal tests

3 Replication



Molecular replication

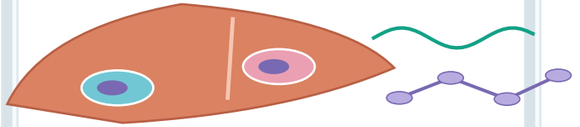
Perez NK • GTEx blood • Nathan T

Disease replication

FinnGen • UK Biobank

Independent evidence only

4 Tissue context



Liver donor-level analysis

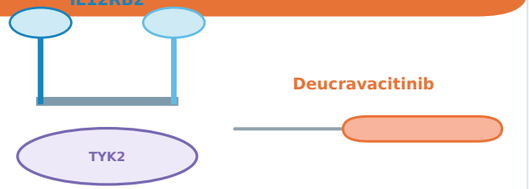
HRA008003: 5 PBC + 5 controls

External liver scRNA • bulk RNA

PXD083842 proteomic audit

Mechanistic context—not a biomarker

5 Translation gate



Direction anchor

TYK2 rs34536443

CONDITIONAL GO

Direct IL12RB2 docking: NO-GO

No direct PBC efficacy trial; no validated IL12RB2 small-molecule pocket/activity series

Research priority—not treatment advice

EVIDENCE BOUNDARIES

Negative and discordant results remain visible in the evidence chain

Liver RNA

No transcriptome-wide FDR-significant case–control contrast

Protein / trial

Compartment-specific discordance; ustekinumab 0/20 responders

Interpretation

Counts are screened combinations, not independent discoveries

CS, credible set; eQTL, expression quantitative trait locus; LD, linkage disequilibrium.